

MTH404:DISCRETE AND COMPUTATIONAL MATHEMATICS

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: learn core concepts of mathematical logic for proposition formulation and proof techniques.

CO2 :: describe lattices and its algebraic structures to form partially ordered sets.

CO3 :: apply elementary counting techniques to solve real world problems.

CO4 :: understand matrix algebra, rank, and inverse in basic problem contexts.

CO5 :: apply probability laws and Bayes' theorem to solve real-life problems.

Unit I

Logic and propositional calculus : introduction, propositions and compound statements, basic logical operations, propositions and truth tables, tautologies and contradictions, logical equivalence, algebra of propositions, conditional and biconditional statements, arguments, propositional functions, quantifiers, negation of quantified statements

Unit II

Ordered sets and lattices : relations, types of relations, partial ordered sets, Hasse diagrams of partially ordered sets, well-ordered sets, lattices, bounded lattices, distributive lattices, introduction to boolean algebra, basic definitions, duality, basic theorems, boolean algebras as lattices

Unit III

Techniques of counting : introduction to basic counting principles, mathematical functions, permutations, combinations, the pigeonhole principle

Unit IV

Linear algebra I : matrix, types of matrices, elementary operations of matrices, determinants, properties of determinants, inverse of matrices, rank of a matrix

Unit V

Linear algebra II : solution of a linear system of equations, Cramer's rule, matrix method, eigenvalues and eigenvectors, properties of eigenvalues

Unit VI

Probability : random experiment, sample space, events, the axioms of probability, algebra of events, conditional probability, multiplication theorem of probability, independent events, Bayes theorem and its applications

Text Books:

1. DISCRETE MATHEMATICS (SCHAUM'S OUTLINES) (SIE by SEYMOUR LIPSCHUTZ, MARC LIPSON, VARSHA H. PATIL,, Tata McGraw Hill, India

References:

1. HIGHER ENGINEERING MATHEMATICS by B.S. GREWAL, KHANNA PUBLISHERS

2. DISCRETE MATHEMATICS AND ITS APPLICATIONS by KENNETH H ROSEN, Mc Graw Hill Education